

RIPPLE: CORPUS-ANCHORED GEOMETRY DISTILLATION FOR TEXT EMBEDDINGS

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ABSTRACT

Relational knowledge distillation for text embeddings commonly matches teacher and student similarities among examples that happen to share a mini-batch. This is a poor way to transfer local geometry: after T epochs with batch size B on a corpus of size N , an anchor is compared with only an $O(TB/N)$ fraction of its teacher neighbors, while relations that never co-occur are absent from the empirical objective. We introduce **RIPPLE**, a corpus-anchored geometry-distillation method that chooses comparisons from the teacher rather than from the batch. RIPPLE caches teacher embeddings once, constructs a mutual k -nearest-neighbor transition operator, and distills its neighborhood distributions into a compact student. One-step operator matching provides local-to-global control of longer diffusion distributions; direct supervision at broader times tightens this guarantee. Diffusion-weighted candidate sampling retains most target mass while scoring only $m \ll k$ examples per anchor. Because normalized row matching determines similarity differences but leaves a row-wise additive ambiguity in score space, RIPPLE also supervises teacher-walk-visited rows whose reciprocal overlaps encourage cross-anchor consistency. The complete method uses only final teacher embeddings and discards the teacher after preprocessing. Across nine datasets and three teacher-student pairs, RIPPLE obtains the best mean overall score among the evaluated compact students in all three settings, with its clearest and most consistent gains on semantic textual similarity.

1 INTRODUCTION

Text embedding models support semantic search, clustering, reranking, and retrieval-augmented generation. Their utility depends not only on individual vectors but on the geometry those vectors induce: which examples are neighbors, how sharply they are separated, and how local neighborhoods connect. Strong embedding teachers often learn this structure at substantial inference cost. A compact student is useful only if it inherits that structure rather than merely reproducing isolated outputs.

Relational knowledge distillation appears to target this requirement directly: it matches teacher and student similarities among examples in a mini-batch (Park et al., 2019; Passalis et al., 2021). The comparisons, however, are selected by the data loader rather than by the teacher. For a corpus of N examples, an anchor sees only $B - 1$ other examples per epoch, almost none of which are among its teacher-defined semantic neighbors. After T epochs, only an $O(TB/N)$ fraction of those neighbors has ever co-occurred with the anchor. The problem is not merely that the resulting estimate is noisy. Similarities to examples that never co-occur do not appear in the empirical objective at all, so the objective does not directly identify the local teacher relations it is meant to transfer.

We propose **RIPPLE**, a *corpus-anchored geometry-distillation* method. RIPPLE first caches the teacher embeddings and constructs a mutual k -nearest-neighbor transition operator over the entire distillation corpus. Training comparisons are then selected from this teacher operator, not inherited accidentally from a mini-batch. The student matches the teacher’s one-step neighborhood distribution and a small family of longer diffusion distributions on sampled candidate sets. This turns

diffusion into a transfer mechanism: accurate one-step matching controls the error of every longer walk, while direct supervision at broader times tightens the resulting bound.

Two additional pieces make operator matching practical and coherent. First, diffusion-weighted sampling covers most of a target row with a candidate budget governed by the target’s concentration rather than by the corpus size; candidates from all anchors are pooled and encoded once per batch. Second, row-wise softmax matching identifies similarity differences within a row but is invariant to an additive row offset in score space. RIPPLE therefore uses teacher random walks to select additional rows for supervision. Reciprocal overlaps between neighboring supervised rows, together with the symmetry of cosine similarity, encourage their offsets to agree and make the learned rows more consistent across anchors. The teacher is discarded after preprocessing, and student training uses no teacher hidden states, online teacher passes, or task labels.

Our contributions are:

- We formalize the coverage and objective-level identification limits of mini-batch relational distillation: after T epochs, an anchor has been compared with at most an $O(TB/N)$ fraction of its teacher neighbors, and unseen relations are absent from the empirical score-level objective (Section 3).
- We introduce RIPPLE, which distills a corpus-level teacher neighborhood operator through one-step and multi-scale diffusion targets, efficient mass-preserving candidate sampling, and overlapping-row supervision for cross-anchor consistency (Section 4).
- Across nine datasets and three teacher–student pairs, RIPPLE obtains the best mean overall score among the evaluated compact students in all three settings, with the clearest gains on semantic textual similarity. Controlled studies are designed to test graph breadth, multi-scale supervision, row supervision, and sampling efficiency separately (Section 5.2).

2 RELATED WORK

Embedding distillation. Knowledge distillation transfers the behavior of a high-capacity teacher to a compact student (Hinton et al., 2015). Language-model compression has used logits, hidden states, attention maps, and intermediate representations (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020; Xu et al., 2024). Embedding-specific methods instead target the geometry used by retrieval and similarity tasks. DistillCSE transfers contrastive sentence-embedding behavior (Gao et al., 2021; Xu et al., 2023); EmbedDistill aligns dense-retrieval representations (Kim et al., 2023); and recent systems such as Gecko, Jasper–Stella, LEAF, and Jina Embeddings v5 combine teacher-generated data, representation matching, or task-specific training (Lee et al., 2024; Zhang et al., 2024; Vujanic & Rueckstiess, 2025; Akram et al., 2026). These methods establish strong embedding students, but they do not directly address which corpus relations an embedding objective exposes to the student.

Relational and neighborhood distillation. Relational KD replaces coordinate matching with relations among examples. RKD matches distances and angles (Park et al., 2019), while PKT matches conditional affinity distributions in representation space (Passalis et al., 2021). In their scalable forms, these distributions or relation matrices are estimated from stochastic mini-batches. Consequently, the objective can preserve the relations present in a batch without ensuring that an anchor’s teacher neighbors were included. Recent neighborhood methods make the selection more explicit. NRKD selects nearest neighbors inside the mini-batch and transfers feature- and logit-level neighborhood relations (Gou et al., 2025). BINGO constructs teacher-selected bags of related instances for self-supervised distillation (Xu et al., 2022), providing corpus-informed comparisons but no operator or multi-scale target. CoSS instead aligns feature coordinates and feature-space structure without explicitly constructing neighborhood distributions (Singh & Wang, 2024). RIPPLE is closest to this family, but anchors every supervised row to a teacher-only corpus operator and transfers normalized neighborhood distributions at several diffusion times.

Graph and diffusion methods. Diffusion Maps use powers of a Markov operator on a neighborhood graph to expose connectivity at different scales (Coifman & Lafon, 2006); HeatGeo uses heat diffusion to construct a geodesic-preserving embedding (Huguet et al., 2023). Graph knowledge distillation typically assumes an observed graph and transfers predictions or representations between

graph models (Tian et al., 2025; Zhang et al., 2022). RIPPLE uses neither an observed task graph nor a student graph. The graph is induced once from cached teacher embeddings, its diffusion rows serve as supervision for a parametric text encoder, and inference on unseen text requires only the student.

The distinguishing choice is therefore not diffusion alone. RKD, PKT, NRKD, and CoSS derive their comparison support from batch co-occurrence; BINGO uses teacher-selected corpus groups but transfers single-scale instance relations; and graph KD begins from an observed task graph or coupled teacher–student graphs. RIPPLE instead lets a teacher-only corpus operator select both the supervised rows and columns, transfers its normalized neighborhood distributions across scales, and uses overlapping rows to make independently normalized anchors more consistent. The student remains a parametric text encoder and requires neither the operator nor the teacher at inference.

3 WHY MINI-BATCH RELATIONS ARE INSUFFICIENT

Relational distillation compares teacher and student similarities over examples that happen to co-occur in a stochastic mini-batch (Park et al., 2019; Passalis et al., 2021). We separate two limitations of this sampling scheme. First, a batch covers almost none of an anchor’s teacher neighborhood. Second, relations that are never sampled are absent from the empirical score-level objective. The first is a coverage statement; the second is an identification statement. Neither depends on the optimizer. Proofs are in Appendix A.

Setup and notation. Let $\mathcal{D} = \{x_i\}_{i=1}^N$ be an unlabeled distillation corpus with cached unit-normalized teacher embeddings t_i , write $c_{ij}^T = \cos(t_i, t_j)$, and let $\mathcal{N}_k(i)$ be the k nearest neighbors of x_i under c^T . A student encoder S_θ produces embeddings s_i and scores $c_{ij}^S = \cos(s_i, s_j)$. We model each epoch as a uniformly random partition of $\mathcal{D}$ into batches of size B , independently resampled between epochs. At temperature τ_0 , a representative batch-relational objective compares

$$p_i^{T,B}(j) \propto \exp(c_{ij}^T/\tau_0), \quad p_i^{S,B}(j) \propto \exp(c_{ij}^S/\tau_0), \quad j \in \mathcal{B} \setminus \{i\}, \quad (1)$$

through $\mathcal{L}_B = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} D_{\text{KL}}(p_i^{T,B} \| p_i^{S,B})$.

Proposition 1 (Neighborhood coverage). *Fix an anchor i and let $\mathcal{B} \ni i$ be its batch. Then*

$$\mathbb{E} |\mathcal{B} \cap \mathcal{N}_k(i)| = \frac{(B-1)k}{N-1}, \quad (2)$$

$$\Pr[\mathcal{B} \cap \mathcal{N}_k(i) = \emptyset] = \frac{\binom{N-1-k}{B-1}}{\binom{N-1}{B-1}} \geq 1 - \frac{(B-1)k}{N-1}. \quad (3)$$

Across T independently resampled epochs, the expected fraction of the k neighbor pairs that ever co-occur with i is exactly

$$q_T = 1 - \left(1 - \frac{B-1}{N-1}\right)^T \leq \frac{T(B-1)}{N-1} = O\left(\frac{TB}{N}\right). \quad (4)$$

Thus $q_T = o(1)$ whenever $TB = o(N)$. The exposed *fraction* is independent of k : enlarging the nominal neighborhood increases the absolute number of relevant relations, but uniform batches still reveal the same vanishing fraction of them. Coverage alone could still be viewed as a sample-efficiency problem. The next result states what the realized objective can identify from the comparisons it actually contains.

Proposition 2 (Score-level under-identification). *Fix all batches realized over T epochs and let $\mathcal{O}_i^T = \bigcup_{\mathcal{B} \ni i} (\mathcal{B} \setminus \{i\})$ be the columns ever compared with anchor i . (i) A realized batch term is zero if and only if, for every $i \in \mathcal{B}$, there is a constant a_i^B such that $c_{ij}^S = c_{ij}^T + a_i^B$ for all $j \in \mathcal{B} \setminus \{i\}$. (ii) Viewed as a function of its score arguments, the empirical objective depends on row i only through $\{c_{ij}^S : j \in \mathcal{O}_i^T\}$. Any two score arrays that agree on these observed entries have exactly the same empirical loss, regardless of their entries outside $\mathcal{O}_i^T$. (iii) Two realized batch offsets for the same anchor are forced to agree only if the corresponding column sets overlap (or are connected by a chain of such overlaps).*

The proposition concerns the information present in the objective, not the realizability of arbitrary score arrays by cosine embeddings. A parametric encoder shares parameters across examples and therefore assigns definite scores even to unseen pairs. Those scores can generalize well, but their values are supplied indirectly by the student’s inductive bias rather than identified by a teacher comparison. Likewise, cosine symmetry and Gram-matrix constraints may reduce the formal score-space freedom without adding the missing teacher observations.

The perfect score row $c_{ij}^S = c_{ij}^T + a_i$ attains zero loss on every restriction, so the problem is not bias from renormalizing teacher and student on the same batch. It is that the empirical objective observes only the accidental comparison set $\mathcal{O}_i^T$, which Proposition 1 shows excludes nearly all teacher neighbors under practical schedules.

The remedy is to choose comparisons from teacher geometry, rather than letting the batch choose them. Section 4 constructs a corpus-level teacher neighborhood operator, explicitly supervises its rows, and then addresses the approximation and row-consistency questions created by that choice.

4 RIPPLE: CORPUS-ANCHORED GEOMETRY DISTILLATION

RIPPLE replaces relations selected by mini-batch co-occurrence with rows selected from a teacher-induced corpus operator. It then transfers those rows at several geometric scales, approximates them on small candidate sets, and supervises overlapping rows to improve cross-anchor consistency. For distributions p, q on a common finite set, let $\text{TV}(p, q) = \frac{1}{2}\|p - q\|_1$; for a column set C , let $p|_C$ denote restriction to C followed by renormalization. Proofs are in Appendix A.

4.1 TEACHER NEIGHBORHOOD OPERATOR

Let $t_i = T(x_i)$ be the cached, unit-normalized embedding of corpus example x_i . Writing $\mathcal{N}_k^T(i)$ for its top- k neighbors under teacher cosine similarity, RIPPLE retains mutual edges,

$$j \in \mathcal{M}_k^T(i) \iff j \in \mathcal{N}_k^T(i) \text{ and } i \in \mathcal{N}_k^T(j), \quad (5)$$

which suppresses one-directional hub edges. The weighted adjacency and transition operator are

$$W_{ij}^T = \mathbf{1}[j \in \mathcal{M}_k^T(i)] \exp(c_{ij}^T/\tau_i), \quad P^T = (D^T)^{-1}W^T, \quad (6)$$

where D^T is the diagonal degree matrix. RIPPLE fixes a target perplexity ρ rather than a global graph bandwidth. For a row of degree $d_i > \rho$, its bandwidth τ_i is chosen so that

$$H(P_i^T) = - \sum_{j \in \mathcal{M}_k^T(i)} P_{ij}^T \log P_{ij}^T = \log \rho. \quad (7)$$

Thus every sufficiently high-degree transition row has the same effective neighborhood size despite differences in the scale and density of teacher similarities (Hinton & Roweis, 2002; van der Maaten & Hinton, 2008; Vladymyrov & Carreira-Perpiñán, 2013). For non-degenerate rows the entropy is monotone in τ_i , so the solution is obtained by bisection; when $d_i \leq \rho$, the target is clipped just below the maximum attainable entropy and the row is nearly uniform. The solved τ_i values are stored with the graph and reused by the corresponding student and walk readouts. They are derived preprocessing quantities, not tunable model parameters. Moreover, under an affine rescaling $c_{ij}^T \mapsto ac_{ij}^T + b$ with $a > 0$, the solution changes as $\tau_i \mapsto a\tau_i$ and P_i^T is unchanged. Isolated nodes fall back to their directed teacher neighbors. After this preprocessing step, the teacher model is no longer needed.

Each row P_i^T is a teacher-selected conditional neighborhood distribution. Unlike an in-batch target, its support is tied to corpus geometry and is available whenever x_i is sampled as an anchor. We use the lazy operator $\tilde{P}^T = \frac{1}{2}(I + P^T)$ to define longer walks without periodicity.

4.2 MULTI-SCALE DIFFUSION DISTILLATION

The sharpest graph target is the one-step row $p_{i,1}^T = P_i^T$. Broader targets are obtained from the lazy teacher walk,

$$p_{i,r}^T = \delta_i^\top (\tilde{P}^T)^r \Big|_{\mathcal{D} \setminus \{i\}}, \quad r \in \{2, 4\}, \quad (8)$$

where the trivial self column is removed and the remaining mass is renormalized. This construction has the interpretation of diffusion on a sampled manifold (Coifman & Lafon, 2006), but the method and results concern the finite teacher operator directly.

The student does not construct a second graph during training. Its one-step row is read from its cosine scores on the *teacher* support and bandwidth,

$$P_i^S = \text{softmax}_{j \in \mathcal{M}_k^T(i)}(c_{ij}^S / \tau_i). \quad (9)$$

Matching this local object controls the diffusion generated by the student rows on the same support.

Theorem 1 (Local-to-global operator transfer). *Let P^T and P^S be teacher and student transition matrices on the same support. Suppose that, for every row i , one-step matching on a candidate set C_i satisfies $D_{\text{KL}}(P_i^T |_{C_i} \| P_i^S |_{C_i}) \leq \varepsilon$, while $P_i^T(C_i) \geq 1 - \delta_T$ and $P_i^S(C_i) \geq 1 - \delta_S$. Then, for every $r \geq 1$,*

$$\max_i \text{TV}(\delta_i^\top (\tilde{P}^T)^r, \delta_i^\top (\tilde{P}^S)^r) \leq \frac{r}{2} (\sqrt{\varepsilon/2} + \delta_T + \delta_S). \quad (10)$$

Thus one-step supervision is sufficient for conditional local-to-global control, but its worst-case bound grows linearly with diffusion time. RIPPLE additionally asks the same student cosine field to predict the teacher’s broader rows. At scale r , define the full student score readout

$$q_{i,r}^S(j) \propto \exp(c_{ij}^S / \tau_r(i)), \quad j \neq i. \quad (11)$$

Unlike $(\tilde{P}^S)^r$, this is a scale-specific readout of student scores rather than a separately materialized student diffusion. It provides direct supervision at the teacher’s broader neighborhood scale.

Corollary 1 (Direct multi-scale tightening). *If at a scale r the restricted divergence satisfies $D_{\text{KL}}(p_{i,r}^T |_{C_i} \| q_{i,r}^S |_{C_i}) \leq \varepsilon_r$ and the teacher and student score rows place at most δ_T and δ_S mass outside C_i , then*

$$\text{TV}(p_{i,r}^T, q_{i,r}^S) \leq \sqrt{\varepsilon_r/2} + \delta_T + \delta_S, \quad (12)$$

with no propagation factor r .

The corollary does not claim that a softmax readout equals the student’s r -step Markov row. It states the narrower fact used by the objective: direct scale- r supervision avoids relying only on the propagated one-step bound. Section 5.2 tests whether this surrogate also reduces the measured student-operator diffusion error at held-out scales.

Different teacher diffusion times cannot be matched meaningfully by one shared student distribution. For the graph scales we use a common candidate domain C_i and scale-specific resolutions $\tau_r(i) = \sqrt{r} \tau_i$, with $\tau_1(i) = \tau_i$.

Proposition 3 (Collapse under a shared resolution). *Let all graph scales for anchor i share the same student distribution q_i^S on the same support, and let $\sum_r \omega_r = 1$. Then*

$$\sum_r \omega_r D_{\text{KL}}(p_{i,r}^T \| q_i^S) = H\left(\sum_r \omega_r p_{i,r}^T, q_i^S\right) - \sum_r \omega_r H(p_{i,r}^T), \quad (13)$$

so optimization is equivalent to matching the single mixture $\sum_r \omega_r p_{i,r}^T$.

Scale-specific temperatures make the student distributions distinct functions of the same cosine scores, preventing this algebraic collapse. By construction, the $r = 1$ and walk readouts reuse the bandwidth of their teacher transition row, and the ambient target uses the same τ_0 on both sides; Appendix B records the remaining temperature ladder.

4.3 EFFICIENT CANDIDATE SAMPLING

The support of an r -step teacher row can be much larger than k , so scoring every anchor against every reachable corpus example is unnecessary and expensive. RIPPLE samples a small set from the target mass.

Proposition 4 (Mass missed by sampled candidates). *Let p be a target distribution and let C contain m independent draws from p . Then*

$$\mathbb{E}[\text{TV}(p|_C, p)] = \mathbb{E}[p(\bar{C})] \leq \min_{t>0} (e^{-tm} + \text{Tail}_p(t)), \quad \text{Tail}_p(t) = \sum_{j:p_j < t} p_j. \quad (14)$$

If every supported atom has mass at least c/ρ_p , where $\rho_p = e^{H(p)}$ is the effective support of the particular target p , the expected missed mass is at most e^{-cm/ρ_p} .

The relevant budget is therefore the target’s effective support, not the corpus size and not the raw number of nodes reachable by diffusion. In the regime studied here, this permits $m \ll k$. The implementation draws high-mass diffusion candidates without replacement using Gumbel top- k sampling (Efraimidis & Spirakis, 2006; Kool et al., 2019), then adds teacher hard negatives and random negatives. These negatives expose the student to columns outside the retained diffusion mass and empirically control the student leakage term δ_S in Theorem 1; the theorem keeps this quantity as an explicit hypothesis.

For a mini-batch B , candidate sets are pooled into $\mathcal{P}_B = \bigcup_{i \in B} C_i$ and deduplicated. Every pooled text is encoded once, while each anchor can be compared with the shared pool. Graph-scale targets are normalized on the anchor’s own C_i . A dense ambient target supplies teacher similarities for the wider shared pool,

$$p_{i,0}^T(j) = \frac{\exp(c_{ij}^T/\tau_0)}{\sum_{j' \in \mathcal{P}_B \setminus \{i\}} \exp(c_{ij'}^T/\tau_0)}, \quad j \in \mathcal{P}_B \setminus \{i\}. \quad (15)$$

This ambient member prevents graph-external columns from being treated as false zero-mass relations; it is included in the complete geometry loss below but is not itself a diffusion time.

4.4 ROW SUPERVISION FOR CROSS-ANCHOR CONSISTENCY

Matching a normalized row determines relative scores within that row but not its additive origin. This is an ambiguity of the score-level observation map, not a claim that arbitrary shifted rows form a realizable cosine Gram matrix.

Proposition 5 (Row-wise additive ambiguity). *On a fixed column set Ω_i , any row-wise softmax term is invariant under $c_{ij}^S \mapsto c_{ij}^S + a_i$. If teacher and student softmaxes use the same temperature, $|\Omega_i| \geq 2$, and the term is zero, then*

$$c_{ij}^S = c_{ij}^T + a_i, \quad j \in \Omega_i, \quad (16)$$

for some row-dependent constant a_i .

RIPPLE uses teacher random walks to choose additional rows whose supports overlap along teacher edges. From each anchor $i \in B$, it samples M non-backtracking walks of length L ,

$$j_0^{(m)} = i, \quad j_{t+1}^{(m)} \sim P_{j_t^{(m)}}^T, \quad t = 0, \dots, L-1. \quad (17)$$

The anchor row at $t = 0$ is already supervised by the one-step target. For every visited row u at $t \geq 1$, let $\Omega_u^w = \mathcal{P}_B \cap \mathcal{M}_k^T(u)$ and let $\nu_B(u)$ be its normalized visit count. RIPPLE matches the stored teacher row on this overlap:

$$\mathcal{L}_{\text{row}} = \sum_u \nu_B(u) D_{\text{KL}}(P_u^T|_{\Omega_u^w} \parallel \text{softmax}_{v \in \Omega_u^w}(c_{uv}^S/\tau_u)). \quad (18)$$

Visited nodes replace random candidates when possible, so the term largely reuses texts already encoded for the batch.

Proposition 6 (Consistency induced by overlapping rows). *Let $\mathcal{S}$ be the supervised rows and construct a constraint graph $\mathcal{G}$ on $\mathcal{S}$, with an edge $\{u, v\}$ whenever $v \in \Omega_u$, $u \in \Omega_v$, and both corresponding same-temperature row terms vanish. Then the score-level offsets of Proposition 5 satisfy $a_u = a_v$ on every edge. Hence admissible row offsets are constant on each connected component of $\mathcal{G}$, leaving at most one offset per component rather than one per isolated row.*

The mechanism is cosine symmetry: $c_{uv}^S = c_{vu}^S$ and $c_{uv}^T = c_{vu}^T$ force the two row offsets to agree. Consecutive nodes of a walk are mutual graph neighbors and their supervised rows therefore add precisely these reciprocal constraints. At nonzero loss the statement is approximate; Appendix A.8 bounds edge-wise offset disagreement and records the coverage argument for walk-selected rows.

4.5 FULL OBJECTIVE AND COMPLEXITY

Let $\mathcal{R} = \{0, 1, 2, 4\}$, with $\Omega_{i,0} = \mathcal{P}_B \setminus \{i\}$ and $\Omega_{i,r} = C_i$ for $r \geq 1$. The student distribution at each scale is

$$p_{i,r}^S(j) = \frac{\exp(c_{ij}^S/\tau_r(i))}{\sum_{j' \in \Omega_{i,r}} \exp(c_{ij'}^S/\tau_r(i))}, \quad j \in \Omega_{i,r}, \quad (19)$$

where τ_0 is shared by the teacher and student ambient targets, $\tau_1(i) = \tau_i$, and $\tau_r(i) = \sqrt{r} \tau_i$ for $r > 1$. With graph targets restricted to C_i , the anchored geometry loss is

$$\mathcal{L}_{\text{diff}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}} \omega_r D_{\text{KL}}(p_{i,r}^T |_{\Omega_{i,r}} \| p_{i,r}^S). \quad (20)$$

The ambient term $r = 0$ is thus explicit inside $\mathcal{L}_{\text{diff}}$. The complete objective is

$$\boxed{\mathcal{L}_{\text{RIPPLE}} = \mathcal{L}_{\text{diff}} + \lambda \mathcal{L}_{\text{row}}}. \quad (21)$$

Teacher encoding is a one-time cost. The current exact nearest-neighbor implementation computes all corpus cosine similarities in $O(N^2 d_T)$ time using chunked memory, retains $O(Nk)$ graph edges, and stores sparse diffusion rows truncated by a fixed mass tolerance. Approximate nearest-neighbor search could replace the quadratic graph-build step, but is not used in the reported experiments. During student training, a batch encodes at most $|B| + |\mathcal{P}_B| \leq |B|(m+1)$ texts and forms $O(|B||\mathcal{P}_B|d_S)$ anchor-candidate similarity work. The row term reuses the pooled encodings and adds only similarities on visited-row overlaps. No teacher forward pass, teacher hidden state, task label, or additional inference-time component is required; after preprocessing, the teacher is discarded.

5 EXPERIMENTS

Models and data. We evaluate three teacher→student pairs: Qwen3-Embedding-4B → BERT-base (109M), BGE-M3 → MiniLMv2-H768 (66M), and Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M). Following TALAS (Dao et al., 2026), the unlabeled distillation corpus contains approximately 15K sentences sampled evenly from Emotion, WiC, and STS-B. Teacher embeddings are cached once and the teacher is removed from GPU memory before student training.

Evaluation. The nine downstream datasets form three task families: classification with macro-F1 on Banking77, Tweet, and Emotion; pair classification with average precision on MRPC, SciTail, and WiC; and semantic textual similarity with Spearman correlation on SICK, STS12, and STS-B. Emotion, WiC, and STS-B are in-domain; the other six datasets are out-of-domain. We select checkpoints and pair-classification thresholds using validation data and evaluate the test sets once. “Avg.” is the unweighted mean of the nine primary metrics.

Baselines and implementation. We compare with the teacher, the undistilled student, SimCSE-unsup, CDM, DSKD, Jasper–Stella, DistillCSE, EMO, and TALAS, using the corresponding TALAS protocol (Dao et al., 2026). Unless varied by an analysis below, RIPPLE uses $k = 200$, target graph perplexity $\rho = 30$, diffusion scales $\{1, 2, 4\}$ plus the ambient target, $m = 64$ candidates, batches of 32 anchors, and five training epochs. Each transition row solves its own bandwidth through Eq. (7); no fixed graph temperature is tuned. Scale weights are proportional to $1/r$, with the ambient and one-step terms equally weighted before normalization. Row supervision uses four non-backtracking walks of length four, begins in the second epoch, and has $\lambda = 0.5$. Additional implementation details are in Appendix B.

5.1 MAIN RESULTS

Table 1 reports all three teacher–student settings. RIPPLE obtains the best mean overall score among the evaluated compact students in each setting: 78.09 versus 78.04 for Qwen3-Embedding-4B → BERT-base, 77.99 versus 76.64 for BGE-M3 → MiniLMv2-H768, and 74.91 versus 73.46 for Qwen3-Embedding-0.6B → MiniLMv2-H384. The first margin is small relative to the reported run variation, whereas the two MiniLMv2 settings improve by 1.35 and 1.45 points.

Table 1: Results across three teacher $\rightarrow$ student settings on nine datasets. Emotion, WiC, and STS-B (marked with $\star$) are in-domain; the remaining datasets are out-of-domain. Bold and underlined values denote the best and second-best results within each setting, excluding the Teacher and Student base reference rows. TALAS and RIPPLE results are reported as mean $\pm$ standard deviation; the remaining baseline results are from TALAS (Dao et al., 2026). For setting (c), the epoch checkpoint is selected on the validation average.

	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Domain Avg		Avg.
	Banking77	Tweet	Emotion*	MRPC	SciTail	WiC*	SICK	STS12	STS-B*	Avg-In	Avg-Out	Avg.
(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	<u>71.53</u>	68.65	63.37	71.42	65.82	75.45	72.24
Jasper and Stella	89.71	<u>73.83</u>	65.98	<u>85.33</u>	80.99	70.50	73.96	67.90	75.65	70.71	78.62	75.98
DistillCSE	90.94	70.90	60.80	82.86	78.98	68.63	75.12	72.61	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	71.47	79.14	76.58
TALAS	90.44 $\pm$ 0.13	74.62 $\pm$ 0.17	70.87 $\pm$ 0.52	85.90 $\pm$ 0.10	<u>81.78 $\pm$ 0.29</u>	68.84 $\pm$ 0.15	77.56 $\pm$ 0.10	<u>73.12 $\pm$ 0.30</u>	<u>79.24 $\pm$ 0.10</u>	72.98 $\pm$ 0.19	<u>80.57 $\pm$ 0.01</u>	<u>78.04 $\pm$ 0.07</u>
RIPPLE	91.75 $\pm$ 0.17	73.49 $\pm$ 0.45	<u>70.21 $\pm$ 0.57</u>	83.38 $\pm$ 0.05	83.47 $\pm$ 0.18	66.58 $\pm$ 0.06	<u>77.21 $\pm$ 0.13</u>	75.48 $\pm$ 0.19	81.23 $\pm$ 0.09	<u>72.67 $\pm$ 0.14</u>	80.80 $\pm$ 0.04	78.09 $\pm$ 0.04
(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)												
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	71.63	83.83	79.76
Student base	81.84	47.70	71.67	79.67	65.06	60.89	49.46	26.76	24.12	44.24	62.41	56.35
SimCSE-unsup	87.89	71.45	55.65	83.87	76.32	67.96	66.93	59.66	63.97	62.53	74.35	70.41
CDM	89.15	70.48	58.71	84.16	75.91	70.00	68.14	62.74	67.78	65.50	75.10	71.90
DSKD	89.51	70.53	57.27	84.39	75.36	<u>69.96</u>	68.81	66.78	70.05	65.76	75.90	72.52
Jasper and Stella	88.38	74.13	62.81	85.98	80.33	68.38	74.35	66.80	74.55	68.58	78.33	75.08
DistillCSE	90.50	72.70	59.88	84.39	78.48	69.49	73.97	70.00	73.73	67.70	78.34	74.79
EMO	90.19	74.26	61.48	85.31	80.82	68.30	75.41	67.06	76.35	68.71	78.84	75.46
TALAS	89.72 $\pm$ 0.19	75.38 $\pm$ 0.11	<u>65.83 $\pm$ 0.58</u>	87.58 $\pm$ 0.02	<u>82.90 $\pm$ 0.12</u>	66.96 $\pm$ 0.07	<u>75.82 $\pm$ 0.11</u>	68.11 $\pm$ 0.20	<u>77.51 $\pm$ 0.18</u>	<u>70.10 $\pm$ 0.13</u>	<u>79.92 $\pm$ 0.05</u>	<u>76.64 $\pm$ 0.04</u>
RIPPLE	90.58 $\pm$ 0.12	73.97 $\pm$ 0.30	71.06 $\pm$ 0.98	85.22 $\pm$ 0.09	83.92 $\pm$ 0.14	62.87 $\pm$ 0.15	78.81 $\pm$ 0.09	73.99 $\pm$ 0.20	81.46 $\pm$ 0.22	71.80 $\pm$ 0.31	81.08 $\pm$ 0.07	77.99 $\pm$ 0.13
(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)												
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	72.66	82.61	79.29
Student base	68.29	41.07	69.90	78.39	64.56	59.58	47.60	21.95	22.08	40.91	58.45	52.60
SimCSE-unsup	86.67	69.56	54.00	82.29	73.76	65.71	66.04	59.11	62.08	60.60	72.91	68.80
CDM	86.07	70.32	53.04	81.97	72.97	68.34	65.97	61.12	65.75	62.38	73.07	69.51
DSKD	85.66	69.86	52.25	82.17	73.52	<u>68.58</u>	66.54	62.52	66.70	62.51	73.38	69.76
Jasper and Stella	85.94	71.25	57.99	83.36	76.17	68.25	70.57	63.81	70.11	65.45	75.18	71.94
DistillCSE	88.03	69.87	54.24	83.20	77.66	67.88	70.09	68.19	71.53	64.55	76.17	72.30
EMO	87.03	<u>71.67</u>	59.63	83.45	78.17	68.89	70.81	65.78	71.42	66.65	76.15	72.98
TALAS	84.16 $\pm$ 0.15	72.82 $\pm$ 0.11	<u>61.13 $\pm$ 0.52</u>	84.85 $\pm$ 0.10	80.91 $\pm$ 0.27	66.32 $\pm$ 0.07	70.72 $\pm$ 0.15	67.80 $\pm$ 0.08	<u>72.39 $\pm$ 0.31</u>	66.61 $\pm$ 0.24	<u>76.88 $\pm$ 0.07</u>	<u>73.46 $\pm$ 0.10</u>
RIPPLE	<u>87.50 $\pm$ 0.24</u>	70.50 $\pm$ 0.29	66.40 $\pm$ 0.40	81.68 $\pm$ 0.12	<u>80.73 $\pm$ 0.05</u>	65.34 $\pm$ 0.10	73.69 $\pm$ 0.11	72.08 $\pm$ 0.09	76.29 $\pm$ 0.08	69.34 $\pm$ 0.13	77.70 $\pm$ 0.11	74.91 $\pm$ 0.08

The most consistent gains occur on semantic textual similarity. RIPPLE gives the best compact-student result on STS-B in all three settings, improving over the strongest baseline by 1.99, 3.95, and 3.90 Spearman points. It leads all three STS datasets in both MiniLMv2 settings. Pair classification is less consistent, and WiC decreases in every setting; the mechanism-specific studies below are therefore necessary to separate improvements in neighborhood geometry from task-family trade-offs.

5.2 BREADTH AS AN INTERVENTION: WHERE DOES ALIGNMENT TRAVEL?

The graph degree k is an intervention on how much teacher geometry is exposed to each anchor. Every run remains corpus-anchored and *only k changes*; candidate budget, scales, row weight, optimizer, and schedule are fixed. We first measure the ambient teacher mass retained by the mutual neighborhood,

$$\text{Mass@}k = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} p_{i,0}^T(\mathcal{M}_k^T(i)), \quad (22)$$

on a fixed evaluation-anchor set $\mathcal{A}$. Mass alone grows almost by construction, so it is not the result of interest. The diagnostic is *where* student alignment improves. We rank the corpus by teacher similarity and report teacher-student Jensen-Shannon divergence on the fixed bands 1:50, 51:200, and 201:800, as well as over the full corpus. A gain confined to ranks already included by k indicates support fitting; a gain in a band before it enters the supervised neighborhood is evidence of local-to-global transfer. The downstream average tests whether either effect matters beyond operator imitation.

Table 2: Neighborhood-breadth intervention. Only k changes. Rank-band and full-corpus columns report teacher-student JS divergence. Off-support improvements are the local-to-global signature. Results are pending.

k	Mass@ k $\uparrow$	JS _{1:50} $\downarrow$	JS _{51:200} $\downarrow$	JS _{201:800} $\downarrow$	Full JS $\downarrow$	Avg. $\uparrow$
50	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
100	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
200	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
400	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

This design admits three distinct outcomes. Monotone in-band gains but flat outer bands mean that anchoring helps only by exposing more pairs. Outer-band gains at small k support transfer through the operator. A late reversal at large k would expose semantic shortcuts from overly broad neighborhoods rather than support the claim that more neighbors are always better.

5.3 LOCAL CONTROL OR MEMORIZED SCALES?

Theorem 1 predicts propagation of one-step error, while direct scale targets should tighten longer-range alignment; Proposition 3 predicts that a shared resolution cannot identify distinct scales. We compare one-step, shared-temperature, and scale-specific supervision with ambient and row terms fixed.

For a fixed set of evaluation anchors, we construct P^S from the trained student on the teacher support using Eq. (9), and report

$$E_r = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \text{TV}(\delta_i^\top (\tilde{P}^T)^r, \delta_i^\top (\tilde{P}^S)^r), \quad r \in \{1, 4, 8, 16\}. \quad (23)$$

Scales 8 and 16 are never supervised, distinguishing transfer from memorization. On a common evaluation pool, E_{sep} measures error in the pairwise separations among the $r \in \{1, 2, 4\}$ readouts; it is small only when the student preserves how distinguishable the teacher scales are. Appendix B.6 gives the definition.

Table 3: Multi-scale mechanism test. E_8, E_{16} are held out; E_{sep} tests scale distinguishability. Results are pending.

Graph-scale supervision	E_1 $\downarrow$	E_8 $\downarrow$	E_{16} $\downarrow$	E_{sep} $\downarrow$	Avg. $\uparrow$
$r = 1$ only	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
$r \in \{1, 2, 4\}$, shared τ	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
$r \in \{1, 2, 4\}$, scale-specific	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

Support requires scale-specific supervision to lower held-out E_8/E_{16} and E_{sep} without trading away E_1 ; improvement only at trained readouts is not operator transfer.

5.4 IS ROW SUPERVISION PRACTICALLY NECESSARY?

Only λ varies in $\mathcal{L}_{\text{RIPPLE}} = \mathcal{L}_{\text{diff}} + \lambda \mathcal{L}_{\text{row}}$, and the sole outcome is downstream Avg. over seeds and settings. A positive- λ lift supports utility; a flat curve falsifies it, while a monotone curve points to extra coverage rather than balanced constraints. Accuracy alone cannot identify calibration. Per-setting curves are reserved in Appendix B.7.

Table 4: Row-loss test. Only λ changes; entries are downstream Avg. (mean $\pm$ standard deviation). Results are pending.

Setting $\backslash \lambda$	0	0.1	0.3	0.5	1.0
Mean over three settings	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

5.5 DOES TARGET MASS PREDICT SAMPLING FIDELITY?

A mass curve alone restates Proposition 4, so Appendix B.8 also compares sampled and full-neighborhood gradients. Only m changes. Support requires mass and gradient fidelity to saturate while $m \ll k$, without downstream loss.

6 LIMITATIONS AND CONCLUSION

RIPPLE inherits teacher-graph errors and domain bias, and its operator is corpus-dependent. Exact construction costs $O(N^2 d_T)$ before sparsification and requires approximate search at scale. Theorem 1 is conditional on local error and omitted mass; student leakage remains empirical.

RIPPLE nevertheless replaces random batch relations with a teacher-anchored operator: one-step matching gives local-to-global control, broader targets supervise coarser scales, sampling provides efficiency, and overlapping rows improve consistency. The teacher is discarded before training.

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A PROOFS

Throughout, $\|v\|_1 = \sum_j |v_j|$ and $\text{TV}(p, q) = \frac{1}{2}\|p - q\|_1$. For a nonnegative vector v with $\|v\|_1 > 0$ and a subset C of its support, $v|_C$ denotes restriction to C followed by renormalization.

A.1 PROOF OF PROPOSITION 1 (NEIGHBORHOOD COVERAGE)

Within an epoch the corpus is partitioned uniformly at random into batches of size B . By exchangeability, the batch containing i contains $B - 1$ of the remaining $N - 1$ examples, uniformly without replacement, so for any fixed $j \neq i$,

$$\Pr[j \in \mathcal{B}] = \frac{B - 1}{N - 1}. \quad (24)$$

The expected count follows by linearity over the k neighbors. The event of seeing no neighbor means that all $B - 1$ batch partners are drawn from the $N - 1 - k$ non-neighbors, giving

$$\Pr[\mathcal{B} \cap \mathcal{N}_k(i) = \emptyset] = \frac{\binom{N-1-k}{B-1}}{\binom{N-1}{B-1}}. \quad (25)$$

The stated lower bound follows from the union bound $\Pr[\mathcal{B} \cap \mathcal{N}_k(i) \neq \emptyset] \leq (B - 1)k/(N - 1)$. Across epochs, let $p_e = (B - 1)/(N - 1)$. A fixed neighbor pair is observed at least once with probability $1 - (1 - p_e)^T$. By linearity this is also the expected observed fraction of the k neighbor pairs, and Bernoulli’s inequality gives $1 - (1 - p_e)^T \leq Tp_e$. $\square$

A.2 PROOF OF PROPOSITION 2 (UNDER-DETERMINATION)

(i) Fix a realized batch $\mathcal{B}$ and $i \in \mathcal{B}$. The divergence in $\mathcal{L}_{\mathcal{B}}$ vanishes exactly when the two distributions of Eq. (1) coincide on $\mathcal{B} \setminus \{i\}$. Taking log-ratios of any two columns j, j' of that set,

$$\frac{c_{ij}^S - c_{ij'}^S}{\tau_0} = \log \frac{p_i^{T, \mathcal{B}}(j)}{p_i^{T, \mathcal{B}}(j')} = \frac{c_{ij}^T - c_{ij'}^T}{\tau_0}, \quad (26)$$

so $c_{ij}^S - c_{ij'}^T$ is constant in j on $\mathcal{B} \setminus \{i\}$; call it $a_i^{\mathcal{B}}$. Conversely, if $c_{ij}^S = c_{ij}^T + a_i^{\mathcal{B}}$ on that set then the factor $e^{a_i^{\mathcal{B}}/\tau_0}$ cancels in the normalization and the two distributions coincide, so the divergence is zero.

(ii) Summing the realized batch terms introduces no additional score arguments: row i occurs only at columns in $\mathcal{O}_i^T$. Therefore two score arrays that agree on every realized argument give the same value to every batch term and hence to the empirical objective, irrespective of entries outside $\mathcal{O}_i^T$.

(iii) If two batches $\mathcal{B}, \mathcal{B}'$ containing i share a column j , then both offsets equal $c_{ij}^S - c_{ij}^T$ and must coincide. Repeating the argument along a chain of overlapping column sets identifies all offsets in that overlap component. With no such chain, the score-level loss contains no equality relating the two constants. This proves the proposition. $\square$

Remark 1 (The defect is identifiability, not bias). The in-batch teacher target $p_i^{T,B}$ is a poor estimator of the teacher’s ambient distribution over the corpus, but this does not by itself impugn the objective: the student is restricted to the same columns, so by part (i) the exact solution $c^S = c^T + a_i$ drives the loss to zero on every batch and the two restrictions cancel. The failure is that almost nothing is constrained, which Eq. (4) quantifies.

A.3 TWO LEMMAS BEHIND THEOREM 1

The theorem combines a perturbation bound, which propagates a one-step error to r steps, with a restriction bound, which converts control on a column set into control of the full row.

Lemma 1 (One step to r steps). *Let P, Q be row-stochastic matrices with lazy walks $\tilde{P}, \tilde{Q}$, and suppose $\max_i \text{TV}(P_i, Q_i) \leq \varepsilon$. Then for every $r \geq 1$, $\max_i \text{TV}(\delta_i^\top \tilde{P}^r, \delta_i^\top \tilde{Q}^r) \leq r\varepsilon/2$.*

Proof. Write $\tilde{P} - \tilde{Q} = \frac{1}{2}(P - Q)$ and telescope:

$$\tilde{P}^r - \tilde{Q}^r = \sum_{s=0}^{r-1} \tilde{P}^s (\tilde{P} - \tilde{Q}) \tilde{Q}^{r-1-s}. \quad (27)$$

Fix a row i and a summand s . The vector $\mu = \delta_i^\top \tilde{P}^s$ is a probability vector, so $\mu(\tilde{P} - \tilde{Q})$ is a convex combination of the rows of $\tilde{P} - \tilde{Q}$ and

$$\|\mu(\tilde{P} - \tilde{Q})\|_1 \leq \max_j \left\| \frac{1}{2}(P_j - Q_j) \right\|_1 = \max_j \text{TV}(P_j, Q_j) \leq \varepsilon. \quad (28)$$

Right-multiplication by a row-stochastic matrix does not increase the ℓ_1 norm: $\|\nu \tilde{Q}^{r-1-s}\|_1 \leq \|\nu\|_1$. Summing the r terms gives $\|\delta_i^\top (\tilde{P}^r - \tilde{Q}^r)\|_1 \leq r\varepsilon$, i.e. $\text{TV} \leq r\varepsilon/2$. $\square$

Lemma 2 (Restriction). *Let p, q be distributions on a common finite set and C a subset with $p(C) \geq 1 - \delta_T$ and $q(C) \geq 1 - \delta_S$. Then $\text{TV}(p, q) \leq \text{TV}(p|_C, q|_C) + \delta_T + \delta_S$.*

Proof. Split the ℓ_1 distance over C and $\bar{C}$. Outside C , $\sum_{\bar{C}} |p_j - q_j| \leq p(\bar{C}) + q(\bar{C}) \leq \delta_T + \delta_S$. Inside C , write $p_j = p(C) p|_C(j)$ and $q_j = q(C) q|_C(j)$:

$$\sum_C |p_j - q_j| \leq p(C) \sum_C |p|_C(j) - q|_C(j)| + |p(C) - q(C)| \leq 2 \text{TV}(p|_C, q|_C) + \delta_T + \delta_S. \quad (29)$$

Adding the two parts and halving yields the claim. $\square$

A.4 LEAKAGE CONTROL FROM NEGATIVE CANDIDATES

The teacher-side defect δ_T in Theorem 1 is a property of the candidate construction; the student-side defect δ_S is what the negatives of Section 4.3 target. Let $\tilde{p} = p|_C$ and $\tilde{q} = q|_C$ be the restricted teacher and student distributions on $C = C^+ \cup C^-$, and suppose $D_{\text{KL}}(\tilde{p} \| \tilde{q}) \leq \varepsilon$ with $\tilde{p}(C^-) \leq \varepsilon_{\text{neg}}$. The partition induces binary distributions $(\tilde{p}(C^+), \tilde{p}(C^-))$ and $(\tilde{q}(C^+), \tilde{q}(C^-))$, and by the data-processing inequality

$$D_{\text{KL}}(\tilde{p} \| \tilde{q}) \geq d_{\text{KL}}(\tilde{p}(C^+) \| \tilde{q}(C^+)), \quad (30)$$

where d_{KL} is the binary divergence. Pinsker’s inequality for the binary pair gives $|\tilde{p}(C^+) - \tilde{q}(C^+)| \leq \sqrt{\varepsilon/2}$, hence $\tilde{q}(C^-) \leq \tilde{p}(C^-) + \sqrt{\varepsilon/2} \leq \varepsilon_{\text{neg}} + \sqrt{\varepsilon/2}$. $\square$

Remark 2 (From sampled negatives to the full complement). The bound above controls the student’s mass on the negative columns actually drawn. The random negatives are uniform draws from the complement of the diffusion support, redrawn every epoch, so over training the student’s scores are constrained on a fresh uniform sample of the complement at every epoch: by exchangeability, a student that hides a constant fraction of softmax mass on some subset of the complement would place mass on the drawn negatives in expectation proportional to that fraction, contradicting a uniformly small restricted loss. A quantitative uniform-convergence version of this argument (a bound on δ_S in expectation over the negative-sampling process) is deferred to the final version. Theorem 1 accordingly takes δ_S as a hypothesis.

A.5 PROOF OF THEOREM 1 (LOCAL-TO-GLOBAL TRANSFER)

Fix an anchor i . Pinsker’s inequality applied to the restricted divergence gives $\text{TV}(P_i^T|_{C_i}, P_i^S|_{C_i}) \leq \sqrt{\varepsilon/2}$. Lemma 2 with $p = P_i^T$, $q = P_i^S$, and $C = C_i$ then bounds the full one-step rows:

$$\text{TV}(P_i^T, P_i^S) \leq \sqrt{\varepsilon/2} + \delta_T + \delta_S =: \varepsilon_{\text{row}}. \quad (31)$$

Applying Lemma 1 with row-wise bound ε_{row} yields, for every $r \geq 1$, $\max_i \text{TV}(\delta_i^\top (\tilde{P}^T)^r, \delta_i^\top (\tilde{P}^S)^r) \leq r \varepsilon_{\text{row}}/2$. $\square$

For Corollary 1, apply Pinsker directly to $p_{i,r}^T|_{C_i}$ and $q_{i,r}^S|_{C_i}$, then use Lemma 2 to extend the comparison to their full score-row domains. Lemma 1 is not invoked, so no factor of r appears. This proves a direct target–readout bound and does not identify $q_{i,r}^S$ with $(\tilde{P}^S)^r$.

Remark 3 (Average-case variant). Training controls the divergence averaged over anchors rather than its maximum. By Jensen’s inequality, $\frac{1}{N} \sum_i \text{TV}(P_i^T|_{C_i}, P_i^S|_{C_i}) \leq \sqrt{\bar{\varepsilon}/2}$ with $\bar{\varepsilon}$ the average restricted divergence, and the telescoping argument of Lemma 1 can be carried out under any fixed initial distribution over anchors, giving the same bounds for anchor-averaged errors.

A.6 PROOF OF PROPOSITION 3 (COLLAPSE)

With one shared student distribution p_i^S ,

$$\sum_r \omega_r D_{\text{KL}}(p_{i,r}^T \| p_i^S) = \sum_r \omega_r [H(p_{i,r}^T, p_i^S) - H(p_{i,r}^T)] = H\left(\sum_r \omega_r p_{i,r}^T, p_i^S\right) - \sum_r \omega_r H(p_{i,r}^T), \quad (32)$$

using that cross-entropy $H(p, q) = -\sum_j p_j \log q_j$ is linear in its first argument and $\sum_r \omega_r = 1$. The subtracted entropy term does not depend on the student. $\square$

A.7 TRUNCATED WALKS AND PROOF OF PROPOSITION 4

The diffusion targets are computed by a truncated walk; its error is additive across steps.

Lemma 3 (Truncation error). *If the walk is computed with per-step truncation to the highest-mass columns followed by renormalization, dropping mass fraction δ_s at step s , then after r steps the computed distribution $\hat{p}$ satisfies $\text{TV}(\hat{p}, p) \leq \sum_{s \leq r} \delta_s$.*

Proof. First, for any nonnegative vector v with total mass 1 and truncation $R(v)$ that zeroes a set of mass δ and renormalizes,

$$\text{TV}(R(v), v) = \frac{1}{2} \left[\sum_{\text{kept}} \left(\frac{v_j}{1-\delta} - v_j \right) + \sum_{\text{dropped}} v_j \right] = \frac{1}{2} [\delta + \delta] = \delta. \quad (33)$$

Let v_s denote the exact walk after s steps and $\hat{v}_s$ the truncated walk, with $M = \tilde{P}$ the (row-stochastic) step operator and δ_s the mass fraction dropped at step s . Then

$$\text{TV}(\hat{v}_s, v_s) \leq \text{TV}(R(\hat{v}_{s-1}M), \hat{v}_{s-1}M) + \text{TV}(\hat{v}_{s-1}M, v_{s-1}M) \leq \delta_s + \text{TV}(\hat{v}_{s-1}, v_{s-1}), \quad (34)$$

using the triangle inequality and that a stochastic matrix does not increase total variation. Induction over $s \leq r$ gives $\text{TV}(\hat{v}_r, v_r) \leq \sum_{s \leq r} \delta_s$. $\square$

Proof of Proposition 4. The identity $\text{TV}(p|_C, p) = p(\bar{C})$ follows from the first display above with $\delta = p(\bar{C})$. For m independent draws from p ,

$$\mathbb{E}[p(\bar{C})] = \sum_j p_j (1 - p_j)^m. \quad (35)$$

Fix $t > 0$ and split the sum: atoms with $p_j \geq t$ contribute at most $(1 - t)^m \sum_j p_j \leq e^{-tm}$, and atoms with $p_j < t$ contribute at most $\text{Tail}_p(t)$. Minimizing over t gives the bound. If every atom in the support of p carries mass at least c/ρ , take $t = c/\rho$: no atom falls below t , the tail term vanishes, and the bound is $e^{-cm/\rho}$. $\square$

Proposition 7 (Coverage of walk-selected constraints). *Let q be the distribution over mutual-neighbor edges traversed by the walks. After n independent traversals, the expected q -mass of edges never traversed is*

$$\sum_e q_e(1 - q_e)^n \leq \min_{t>0} (e^{-tn} + \text{Tail}_q(t)), \quad \text{Tail}_q(t) = \sum_{e: q_e < t} q_e. \quad (36)$$

Adding traversed edges can only merge components of the realized constraint graph, so its component count is non-increasing with additional traversals.

Proof of Proposition 7. Index the edges of $\mathcal{M}_k^T$ and let q_e be the probability that edge e is traversed in a given optimization step, independently across steps. The probability that e is never traversed in n steps is $(1 - q_e)^n$, so the expected q -mass of untraversed edges is $\sum_e q_e(1 - q_e)^n$, which is the quantity bounded in Proposition 4 with p replaced by q and m by n ; the same split at a threshold t gives the same bound. Adding an edge to the traversed set can only merge two components of $\mathcal{G}$ or leave them unchanged, so κ is non-increasing in n . $\square$

Remark 4 (Sampling without replacement). The implementation draws candidates by Gumbel top- k , which realizes weighted sampling without replacement (Efraimidis & Spirakis, 2006; Kool et al., 2019). Draws are distinct, so after m draws the covered mass is at least that of m independent draws in the natural coupling that re-draws duplicates; the bound of Proposition 4 therefore remains valid.

A.8 PROOFS OF PROPOSITIONS 5 AND 6 (GAUGE AND ITS REDUCTION)

Invariance. For any temperature τ and column set Ω , $\text{softmax}((c_i + a_i)/\tau)_{j \in \Omega} = \text{softmax}(c_i/\tau)_{j \in \Omega}$, since the constant $e^{a_i/\tau}$ cancels between numerator and normalizer. Every term of $\mathcal{L}_{\text{diff}}$ is a divergence between a fixed teacher target and such a softmax, so the loss is unchanged.

Zero set. Suppose the ambient-scale term vanishes for anchor i on the column set Ω , $|\Omega| \geq 2$: then $\text{softmax}(c_i^S/\tau_0) = \text{softmax}(c_i^T/\tau_0)$ on Ω . Taking log-ratios of any two columns $j, j' \in \Omega$,

$$\frac{c_{ij}^S - c_{ij'}^S}{\tau_0} = \log \frac{p_{i,0}^T(j)}{p_{i,0}^T(j')} = \frac{c_{ij}^T - c_{ij'}^T}{\tau_0}, \quad (37)$$

so $c_{ij}^S - c_{ij'}^T$ is constant in j on Ω ; call it a_i . This proves Proposition 5.

Walk rows carry the same form of constraint. The teacher’s transition row is itself a softmax of teacher cosines at that row’s bandwidth, $P_j^T(j') \propto \exp(c_{jj'}^T/\tau_j)$ on $\mathcal{M}_k^T(j)$, and restricting a softmax to a subset and renormalizing is the softmax over that subset. Hence $P_j^T|_{\Omega_j^w} = \text{softmax}_{\Omega_j^w}(c_j^T/\tau_j)$, and the row term of Eq. (18) vanishes at j exactly when $\text{softmax}_{\Omega_j^w}(c_j^S/\tau_w(j)) = \text{softmax}_{\Omega_j^w}(c_j^T/\tau_j)$. With $\tau_w(j) = \tau_j$ the log-ratio argument above applies verbatim on Ω_j^w — the common temperature cancels from both sides of the ratio, so its value is immaterial and only the tie matters — giving $c_{jv}^S = c_{jv}^T + a_j$ for all $v \in \Omega_j^w$. The argument is thus unchanged whether the bandwidths are shared or solved per row.

Symmetry closes the gauge along edges. Every supervised row $u \in \mathcal{S}$ therefore satisfies $c_{uv}^S = c_{uv}^T + a_u$ on Ω_u , whether it was supervised by the ambient term or by the row term. Let $\{u, v\}$ be an edge of $\mathcal{G}$, so $v \in \Omega_u$ and $u \in \Omega_v$. Applying the identity in both directions,

$$c_{uv}^S = c_{uv}^T + a_u, \quad c_{vu}^S = c_{vu}^T + a_v. \quad (38)$$

Cosine similarity is symmetric, $c_{uv}^S = c_{vu}^S$ and $c_{uv}^T = c_{vu}^T$, so subtracting gives $a_u = a_v$. A function constant across every edge is constant on every connected component, so the score-row observation model leaves at most one independent offset per component of $\mathcal{G}$. This is an upper bound on score-level ambiguity; it does not assert that arbitrary shifted score arrays are realizable cosine Gram matrices. This proves Proposition 6. $\square$

Remark 5 (Approximate consistency). Suppose a supervised term at row u has divergence at most ε rather than zero, with teacher target p and student distribution q on Ω_u , both softmaxes at the

same temperature τ . Pinsker gives $\|p - q\|_\infty \leq \text{TV}(p, q) \leq \sqrt{\varepsilon/2} =: \eta$. For any column j with $p_j \geq p_{\min} > \eta$,

$$|\log q_j - \log p_j| = \left| \log \left(1 + \frac{q_j - p_j}{p_j} \right) \right| \leq \frac{\eta}{p_{\min} - \eta}, \quad (39)$$

so for two columns j, j' the log-ratio identity used above holds up to $2\eta/(p_{\min} - \eta)$, giving $|(c_{uj}^S - c_{uj'}^S) - (c_{uj}^T - c_{uj'}^T)| \leq 2\tau\eta/(p_{\min} - \eta)$. Writing a_u for the offset fitted at row u , the same estimate bounds $|a_u - a_v|$ across an edge $\{u, v\}$ of $\mathcal{G}$ by $O(\tau\sqrt{\varepsilon}/p_{\min})$. Along a path of ℓ edges the bounds add, so offsets identified through a long chain are looser than those identified through a short one: coverage of $\mathcal{G}$ improves with walk length while the accuracy of each identification degrades with it.

Remark 6 (The walk lays down the edges). Consecutive nodes j_t, j_{t+1} of a walk with $t \geq 1$ are adjacent in $\mathcal{M}_k^T$, which is symmetric by construction, and both are supervised rows whenever both lie in $\mathcal{P}_B$. Then $j_{t+1} \in \Omega_{j_t}^w$ and $j_t \in \Omega_{j_{t+1}}^w$, so the step is an edge of $\mathcal{G}$: a walk of length L contributes a path of up to $L - 1$ edges. Anchor-only supervision can also create an edge if two realized anchor rows contain one another, but this depends on reciprocal inclusion. The walk deliberately selects adjacent rows and therefore densifies such constraints.

Remark 7 (Downstream tasks are not gauge-invariant). Consider two anchors i, i' and a downstream decision that thresholds cosine similarity at a shared value θ , as in pair classification or a global retrieval cutoff. Offsets $a_i = +\Delta$, $a_{i'} = -\Delta$ leave $\mathcal{L}_{\text{diff}}$ unchanged by Proposition 5, yet move pairs of i above the threshold and pairs of i' below it, changing the decisions. Rank correlations computed across anchors, as in semantic textual similarity, are affected in the same way. The gauge is therefore task-relevant. A single offset shared across a component is the benign case: it shifts every cosine in the component equally, so it preserves all orderings and moves only the absolute location of θ .

B IMPLEMENTATION NOTES AND ANALYSIS PROTOCOLS

This appendix records graph-bandwidth calibration, the temperature ladder, and the walk and analysis details deferred from Section 4. The implementation has no separate teacher and student temperature knobs for transition-row or ambient matching: each such term reuses the temperature with which its target was constructed.

B.1 ENTROPIC ROW-BANDWIDTH CALIBRATION

For one mutual-neighborhood row, write $p_j(\beta) \propto \exp(\beta c_{ij}^T)$ with inverse bandwidth $\beta = 1/\tau_i$. Its entropy satisfies

$$H_i(\beta) = \log Z_i(\beta) - \beta \mathbb{E}_{p(\beta)}[c_{ij}^T], \quad \frac{dH_i}{d\beta} = -\beta \text{Var}_{p(\beta)}[c_{ij}^T]. \quad (40)$$

Hence H_i decreases monotonically with β and, for a non-degenerate row whenever the requested entropy is attainable, Eq. (7) has a unique positive solution. We find it by bracketing and bisection. A row with degree $d_i \leq \rho$ cannot attain perplexity ρ ; its target entropy is clipped to $(1 - 10^{-3}) \log d_i$ to keep the numerical solution finite. A degree-one or constant-score row is assigned its only meaningful uniform distribution. We store the resulting τ_i with each transition row and report the number of clipped rows during graph construction.

This calibration also removes teacher-specific similarity scale from the graph distribution. If $c'_{ij} = ac_{ij}^T + b$ for $a > 0$, then $\tau'_i = a\tau_i$ yields exactly the same normalized row. The experimental hyperparameter is therefore the effective neighborhood size $\rho = 30$, not a corpus-wide temperature.

B.2 THE REST OF THE TEMPERATURE LADDER

The $r = 1$ readout uses the graph bandwidth, $\tau_1(i) = \tau_i$. Broader scales are set by $\tau_r(i) = \sqrt{r} \tau_i$, following the $\sqrt{t}$ spread of a diffusion at nominal time r . This is a rule rather than a derivation: the lazy walk takes $r/2$ real steps in expectation, so a strict derivation would carry a different constant. The ladder therefore has no free entries and automatically inherits entropic per-row bandwidths. Scale weights follow the complementary rule $\omega_r = 1/r$ with $\omega_0 = \omega_1$, so influence falls with diffusion time.

B.3 TRUNCATION TOLERANCE

Sparse diffusion, the per-anchor candidate pool, and the walk support all perform the same operation: keep a subset of a probability row and renormalize. RIPPLE fixes one tolerance δ and keeps, per row, the smallest column set carrying $1 - \delta$ of the mass. Such a truncation perturbs the row by exactly δ in total variation and by $-\log(1 - \delta)$ nats in Kullback–Leibler divergence—a bound in the units of the loss itself—and by Lemma 3 the error is additive across steps, so an r -step target is within $r\delta$ of the exact one. Every stored array width is therefore a consequence of δ . Memory guards cap the widest rows; the build reports if either bound binds before the tolerance is met, in which case the guarantee does not hold.

B.4 NON-BACKTRACKING WALKS

The walk of Eq. (17) is non-backtracking: the step at $t + 1$ is drawn from $P_{j_t}^T$ conditioned on $j_{t+1} \neq j_{t-1}$, except at a node of degree one, where the edge just traversed is the only one available. Since rows are supervised with weight equal to their visit count, an excursion that returns immediately spends the walk budget re-supervising a row it has already seen; forbidding it raises the number of distinct rows a walk of given length reaches, and non-backtracking walks also have faster mixing and cover time on graphs of bounded degree (Alon et al., 2007). The constraint changes only the occupancy measure ν_B —which rows are chosen—and never a target: a visited node j is still matched against its own transition row P_j^T at τ_j . Proposition 6 is likewise unaffected, since it counts *traversed edges* and a repeated edge contributes no new constraint.

B.5 DENSE RATHER THAN SAMPLED WALK TARGETS

A natural alternative scores the sampled transitions themselves, $-\log p^S(j_{t+1} | j_t)$ summed along each path. The walk is first-order and each step is scored on the teacher’s own j_t , so that path likelihood factorizes into one-step terms whose expectation over the walk is exactly Eq. (18): the sampled form is an unbiased but higher-variance estimator of the dense one. Since P_j^T is already materialized during graph construction, we use the dense form and pay no variance. Nothing in the term couples two consecutive steps; it matches one-step transition rows at walk-visited nodes.

B.6 SCALE-SEPARATION DIAGNOSTIC

On a common candidate pool for every scale, the diagnostic in Table 3 is

$$E_{\text{sep}} = \frac{1}{|\mathcal{A}| \binom{|\mathcal{R}|}{2}} \sum_{i \in \mathcal{A}} \sum_{r < s} |\text{JS}(p_{i,r}^T, p_{i,s}^T) - \text{JS}(q_{i,r}^S, q_{i,s}^S)|, \quad \mathcal{R} = \{1, 2, 4\}. \quad (41)$$

Using a common pool is essential: otherwise changing support could create apparent separation even when two student readouts assign identical scores to shared columns.

B.7 PER-SETTING ROW-LOSS GRID

Table 5 expands the aggregate in Table 4. Only λ changes and every cell remains the downstream nine-task average; no calibration diagnostic is used for model selection.

Table 5: Per-setting row-loss sweep. Entries are downstream Avg. (mean $\pm$ standard deviation). Results are pending.

Setting $\setminus \lambda$	0	0.1	0.3	0.5	1.0
Qwen-4B $\rightarrow$ BERT	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
BGE-M3 $\rightarrow$ MiniLM-768	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Qwen-0.6B $\rightarrow$ MiniLM-384	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

B.8 SAMPLING-EFFICIENCY MEASUREMENT PROTOCOL

For a fixed probe batch, define the gradient fidelity of an m -candidate objective as

$$G_m = \cos\left(\nabla_{\theta}\mathcal{L}_{\text{diff}}^{(m)}, \nabla_{\theta}\mathcal{L}_{\text{diff}}^{(\text{full})}\right). \quad (42)$$

Only m changes across Table 6; the sampler and all other training choices are fixed. Target mass is measured before renormalizing the sampled row. G_m is computed at the same checkpoint and on the same probe batch for every m ; the full-support gradient is used only for evaluation. Step time excludes graph preprocessing and is averaged after warm-up, while peak accelerator memory is reset immediately before the measured steps. All timing and memory results use the same hardware and software stack.

Table 6: Candidate-budget mechanism test. Only m changes. Results are pending.

m	Target mass $\uparrow$	$G_m \uparrow$	Avg. $\uparrow$	Time/step $\downarrow$	Memory $\downarrow$
16	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
32	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
64	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
128	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
256	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]